

BIOLOGY 101 · WEEK 1 QUIZ

Biology 101 — Week 1

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **6 of 8** right to earn the badge.

1. What is a cell?☐

A. The smallest living unit of a body, with its own instruction book inside.

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B. The smallest building block of all matter, whether living or not.

☐

C. A tiny air-filled room inside the lungs where oxygen is stored.

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D. The hard outer covering that protects an organ from damage.

2. When Robert Hooke looked at cork under his microscope in 1665, why did he call the tiny boxes “cells”?☐

A. They rang like tiny bells when he tapped the thin slice of cork.

☐

B. They matched the shape of the lenses he ground for his microscope.

☐

C. They reminded him of the small plain rooms where monks slept.

☐

D. They looked like the prison cells of London's Tower.

3. Which part of the cell works like a power plant, burning sugar with oxygen to make energy?☐

A. Cytoplasm

☐

B. Mitochondria

☐

C. Membrane

☐

D. Nucleus

4. One cell divides in two. Then each new cell divides, and then each of those divides – three rounds of division in total. How many cells are there now?☐

A. 3

☐

B. 8

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C. 4

☐

D. 6

5. What is unusual about a red blood cell compared with most other cells?

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- A. It is the only cell in the body that never wears out or dies.
- B. It is the biggest cell in the body, about the size of a grain of sand.
- C. It has two nuclei, so it can divide twice as fast as other cells.
- D. It has no nucleus, which leaves more room to carry hemoglobin.

6. Which list is in the right order, from smallest to biggest?

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- A. Cell, tissue, organ
- B. Tissue, cell, organ
- C. Organ, tissue, cell
- D. Cell, organ, tissue

7. Thinking like a biologist: why does a very premature baby's skin lose heat and water so easily?

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- A. Its skin cells are much larger than an adult's, so they hold more water to lose.
- B. Its skin has extra layers that trap cold air against the body.
- C. Its skin cells have no membranes yet, so nothing stays inside them.
- D. Its skin has fewer cell layers, so there is less barrier between inside and outside.

8. About how many cells are in an adult human body?

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- A. About 30 trillion
- B. About 30 thousand
- C. About 30 million
- D. About 30 billion

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)